Q12. Prompt Evaluation, Comparison, and Refinement

Aim: To evaluate different prompts, compare their responses, and refine the prompt to improve output quality.

Algorithm:

1. Define a task about the benefits of solar energy.
2. Create three prompts with different levels of detail.
3. Prepare sample responses for each prompt.
4. Evaluate the responses using relevance, accuracy, completeness, clarity, and format.
5. Calculate the average score and select the best refined prompt.

Program:

```
task = "Explain the benefits of solar energy."

prompts = {
    "Prompt A": "Explain solar energy.",
    "Prompt B": (
        "Explain five benefits of solar energy in simple English "
        "using bullet points."
    ),
    "Prompt C": (
        "Explain five benefits of solar energy in simple English. "
        "Use numbered points, include a short example, and conclude "
        "with one sentence about environmental protection."
    )
}

# Sample responses for demonstration
responses = {
    "Prompt A": (
        "Solar energy is renewable and helps reduce electricity costs."
    ),
    "Prompt B": (
        "1. Renewable source.\n"
        "2. Reduces electricity bills.\n"
        "3. Lowers air pollution.\n"
        "4. Requires little maintenance.\n"
        "5. Reduces dependence on fossil fuels."
    ),
    "Prompt C": (
        "1. Renewable: Sunlight is naturally available.\n"
        "2. Saves money: Solar panels can reduce electricity bills.\n"
        "3. Reduces pollution: Solar power produces no direct air "
        "pollution during operation.\n"
        "4. Low maintenance: Panels generally need limited upkeep.\n"
        "5. Energy independence: Solar power can reduce reliance "
```

```
        "Example: A home can use rooftop panels to generate electricit
        "Conclusion: Solar energy helps protect the environment."
    )
}

# Illustrative scores out of 5
scores = {
    "Prompt A": [4, 5, 2, 4, 2],
    "Prompt B": [5, 5, 4, 5, 5],
    "Prompt C": [5, 5, 5, 5, 5]
}

criteria = [
    "Relevance", "Accuracy", "Completeness",
    "Clarity", "Format Adherence"
]

for name in prompts:
    average = sum(scores[name]) / len(scores[name])
    print("\n", name)
    print("Prompt:", prompts[name])
    print("Response:", responses[name])
    print("Scores:", dict(zip(criteria, scores[name])))
    print("Average:", round(average, 2), "/ 5")

best = max(scores, key=lambda name:
           sum(scores[name]) / len(scores[name]))

print("\nBest Refined Prompt:", best)
print("Final Score:",
      round(sum(scores[best]) / len(scores[best]), 2), "/ 5")
print("Prompt evaluation completed successfully!")
```

Output:

```
...
 Prompt A
Prompt: Explain solar energy.
Response: Solar energy is renewable and helps reduce electricity costs.
Scores: {'Relevance': 4, 'Accuracy': 5, 'Completeness': 2, 'Clarity': 4, 'Format Adherence': 2}
Average: 3.4 / 5

 Prompt B
Prompt: Explain five benefits of solar energy in simple English using bullet points.
Response: 1. Renewable source.
2. Reduces electricity bills.
3. Lowers air pollution.
4. Requires little maintenance.
5. Reduces dependence on fossil fuels.
Scores: {'Relevance': 5, 'Accuracy': 5, 'Completeness': 4, 'Clarity': 5, 'Format Adherence': 5}
Average: 4.8 / 5

 Prompt C
Prompt: Explain five benefits of solar energy in simple English. Use numbered points, include a short example, and conclude with one sentence about environmental protection.
Response: 1. Renewable: Sunlight is naturally available.
2. Saves money: Solar panels can reduce electricity bills.
3. Reduces pollution: Solar power produces no direct air pollution during operation.
4. Low maintenance: Panels generally need limited upkeep.
5. Energy independence: Solar power can reduce reliance on grid electricity.
Example: A home can use rooftop panels to generate electricity.
Conclusion: Solar energy helps protect the environment.
Scores: {'Relevance': 5, 'Accuracy': 5, 'Completeness': 5, 'Clarity': 5, 'Format Adherence': 5}
Average: 5.0 / 5

Best Refined Prompt: Prompt C
Final Score: 5.0 / 5
Prompt evaluation completed successfully!
```

Result: Prompt C achieved the highest illustrative score because it provides clear instructions about content, structure, examples, and conclusion.